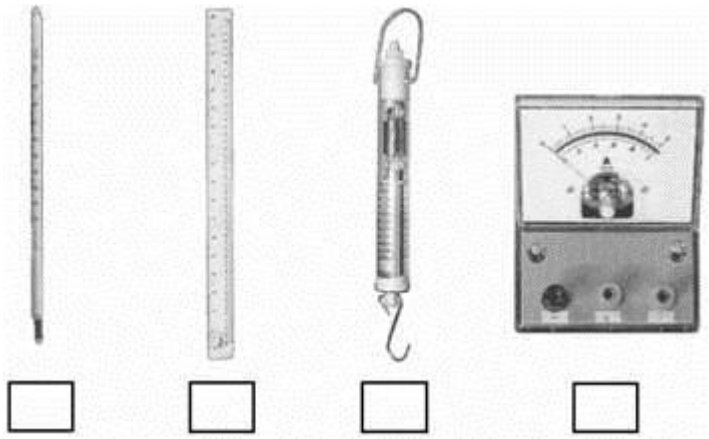
Q1.

Lee blew across the top of paper tubes to make sounds.

He investigated how changing the length of a tube affects the pitch of the sound.

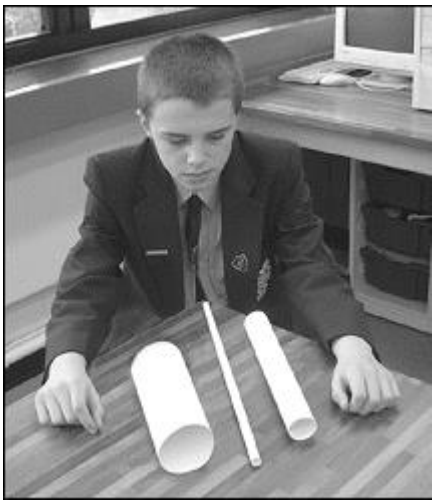
(a) What equipment could he use to measure the length of the tubes?

Tick the correct box.



1 mark

(b) The photograph below shows the different lengths of tubes Lee used.



Suggest **one** way his test might **not** have been fair.

.....

.....

1 mark

(c) Lee made a prediction.

Which of these statements is a prediction?
Tick the correct box.

The tubes were made of paper. ☐

The pitch of the sound is how high or low it is. ☐

The longer tube will make a lower sound. ☐

The sound is caused by the vibration of air. ☐

1 mark

(d) Lee blew across the ends of 3 different lengths of tube and compared the pitch of the sound produced.

His results5 are shown below.

Length of the tube, in cm	*pitch of the sound*
5	*high*
25	*medium*
50	*low*

Which length of tube made the sound with the highest pitch?

..... cm

1 mark
Maximum 4 marks

Q1.

(a) ruler ✓

if more than one box is ticked, award no mark

1 (L3)

(b) any **one** from

- tubes had different widths

  accept 'tubes had different shapes'

- the tubes had different bores
- he blew in different ways
- different thickness of paper
- different paper

  accept 'tubes were different sizes'
  accept 'tubes are one big, one medium, one little'
  *do **not** accept 'different lengths*

1 (L3)

(c) the longer tube will make a lower sound ✓

if more than one box is ticked, award no mark

1 (L4)

(d) 5

1 (L3)

[4]